What a Zero Readout Certifies

Zero as the failure locus of retained distinction

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THE SETTING

Difference is primitive: to read is to tell two things apart and retain a finite record. A **declared structure** g is a finite weighted graph stating which pairs this reader is arranged to compare. A field Φ assigns a rational to each vertex. The retained information of Φ is

$$I_g(\Phi) \;=\; \sum_{(i,j,w)\in g} w\,(\Phi_i - \Phi_j)^2 \;=\; \Phi^\top L_R\,\Phi, \qquad L_R = D_W - W.$$

Everything is over $\mathbb{Q}$. No limit, no completion, no real line.

THE THEOREM

Let every weight of g be **strictly positive**. Then for every field Φ:

$$I_g(\Phi) = 0_{\mathbb{Q}} \iff \Phi_i = \Phi_j \text{ on every edge} \iff \Phi \text{ is constant on every component.}$$

THE PROOF, COMPLETE

1. Each term $w(\Phi_i - \Phi_j)^2$ is a nonnegative weight times a square, hence ≥ 0.
2. A sum of nonnegatives is 0 only if every term is 0. *(induction on the edge list)*
3. $w\,d^2 = 0$ with $w > 0$ forces $d^2 = 0$, and $d^2 = 0$ forces $d = 0$. *($\mathbb{Q}$ is an integral domain)*
4. Equality propagates along paths; conversely a constant field kills every term.

Strict positivity is spent exactly once, at step 3. One edge of weight 0 breaks the theorem: it contributes nothing while its endpoints differ.

TWO EXAMPLES THAT FIX THE MEANING

Zero on a field that is nowhere zero. Let $\Phi_i = 17$ everywhere on a connected g. Then $I_g(\Phi) = 0_{\mathbb{Q}}$. Nothing is empty, no value is zero, nothing is absent — what is absent is any *difference* for the structure to retain.

Zero across a gap in the declaration. Let g have components A, B with $\Phi = 0$ on A and $\Phi = 17$ on B. Then $I_g(\Phi) = 0_{\mathbb{Q}}$ although Φ is not constant. Where g supplies no path it supplies no comparison; the reader was never arranged to tell those apart.

WHAT THE THEOREM LICENSES

Two objects, marked apart, because conflating them is a category mistake:

$$0_{\mathbb{Q}} \in \mathbb{Q} \;\; \text{(a value in the } \textit{codomain}\text{)} \qquad\qquad \mathbf{0}_g := I_g^{-1}(\{0_{\mathbb{Q}}\}) \;\; \text{(its fibre in the } \textit{domain}\text{)}$$

Claim. For this operator under the stated hypotheses, a readout returning $0_{\mathbb{Q}}$ certifies membership in $\mathbf{0}_g$: *the class of fields across which the declared structure retains no difference.* Exact in both directions, by theorem — not by definition.

Not claimed. That $0_{\mathbb{Q}}$ fails to be a mathematical object — it is a rational, and in a generated tower also the ground of $\mathbb{N}$. That this displaces any account of arithmetic zero — it answers a different question. That the operator is faithful *absolutely* — example two shows it is faithful relative to g, which is the strongest form available to a reader with a finite declaration.

Also not this. $0_{\mathbb{Q}}$ is a fact about the states: no difference is retained. The unresolved element $\perp$ is a fact about the reader: it cannot tell. Only the first is determinate, which is why only the first admits a biconditional.

STATUS

The mathematics is elementary and classical over $\mathbb{R}$ (Dirichlet energy vanishes iff constant on components; Chung, Godsil–Royle). **No novelty is claimed against it.** What is offered is its placement as a *faithfulness condition*: over $\mathbb{Q}$, with the hypothesis explicit, machine-checked with no additional global assumptions reported for the audited constants, and shown to coincide in both directions with the failure locus of the primitive — closing a join that was previously carried by interpretation.

Verified under Coq 8.20 and Rocq 9.2: the audited constants report *Closed under the global context*. The repository also includes executable witnesses for the strict-positivity and disconnectedness boundaries. Source: github.com/morrocwi/zero-readout-certifies • archived at doi:10.5281/zenodo.21665100. Review the claim matrix, then compile the artifact.

Arithmetic supplies the symbol $0_{\mathbb{Q}}$; under the stated hypotheses, the theorem determines what a zero returned by this operator certifies.